Joy Shah

New Brunswick, NJ - (615) 484-3314 - joypshah2003@gmail.com - linkedin.com/in/joy-shah62 - github.com/joyshah62

EDUCATION

Rutgers University – New Brunswick — **New Brunswick, NJ, USA**
M.S. in Computer Science (GPA: 3.75/4) — *Aug 2025 – May 2027*

Charusat University — **Nadiad, India**
B.Tech. in Computer Science & Engineering (CGPA: 9.15/10) — *Dec 2021 – Jun 2025*

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL
AI & LLM Systems: LLM Agents, Tool Calling, RAG, Hybrid Search, Re-ranking, Prompt Engineering, Gemini, ElevenLabs
Machine Learning: PyTorch, scikit-learn, BERT, YOLO, OpenCV, Pandas, NumPy
Web & Backend: FastAPI, Flask, React, REST APIs, SQLAlchemy, Alembic, JWT, JupyterLab Extensions
Databases: PostgreSQL, pgvector, MySQL, MongoDB
Cloud & DevOps: AWS (S3), Docker, Linux, Nginx, Git, Bash

WORK EXPERIENCE

Software Engineer Intern – CodeBench

Rutgers University, New Brunswick, NJ — *Jun 2026 – present*

- Developed full-stack capabilities for a JupyterLab-based learning platform, integrating React/TypeScript interfaces with Python server extensions, JWT-secured FastAPI services, relational models, and Alembic migrations.
- Engineered a course-aware AI teaching agent with tool-calling workflows for retrieving notebook context, inspecting and modifying cells, testing code, and searching long-term memory while enforcing guided Socratic responses.
- Built observable AI and assessment workflows by persisting conversations, tool calls, and outputs for replay and analysis while supporting content imports, automated grading, and instructor analytics.

Software Engineer Intern, AI/ML Security

Forenzy Networks, Vadodara, India — *Jan 2025 – Apr 2025*

- Engineered a multi-stage web attack detection system using 5 specialized BERT classifiers with rule-based aggregation, achieving a 97.2% F1-score on stratified validation.
- Built an end-to-end Linux security pipeline integrating Nginx, Apache ModSecurity, log parsing, BERT inference, and Flask, processing 2GB+ of HTTP telemetry daily; log normalization cut false positives from 23% to 4%.
- Containerized the system with Docker Compose and developed a real-time Flask/Chart.js dashboard exposing attack distributions, model confidence, and per-model explanations, reducing SOC triage time by 65%.

Machine Learning Research Intern

CyberNGO (Forenzy Networks), Vadodara, India — *May 2024 – Jun 2024*

- Curated and annotated a 100K+ record multi-label fraud dataset spanning SMS, email, and WhatsApp, and built a Python NLP preprocessing pipeline (lemmatization, transliteration) to structure noisy text.
- Developed a TF-IDF logistic regression classifier for multi-class fraud triage and deployed it into an analyst-facing web portal with an admin dashboard, reducing manual review effort.

PROJECTS

Composur – LLM-Powered Mock Interview Platform

- Built a fully voice-driven AI interviewer that adapts every follow-up question to the candidate's resume and previous answers, by integrating ElevenLabs speech-to-text and text-to-speech with Gemini and scoring each session against a weighted rubric.
- Delivered more relevant interview questions via a 3-stage hybrid search pipeline matching by meaning and keywords (Snowflake Arctic-Embed + pgvector), then merging and re-ranking results with an MS-MARCO cross-encoder.
- Enabled live coding rounds in 3 languages (Java, Python, C++) by running each submission in its own dedicated Docker container, isolating untrusted code from the main application.

Facial Liveness Detection for Biometric Anti-Spoofing — ***Published in IEEE CICT 2024***

- Built a Python anti-spoofing system combining YOLOv8 face detection, Dlib landmark extraction, and a KNN classifier, achieving 98.5% accuracy on a combined custom, NUAA, and MSU-MFSD dataset.
- Assembled and annotated the self-collected dataset and added a liveness stage to reject static-image and replay attacks.

ACHIEVEMENTS

- **Winner, Rutgers Health Hack 2026** – Won the Sevaro track for Cerevia, a real-time tele-neurology command center that forecasts staffing gaps and recommends credential-aware physician shifts via deterministic, explainable rules.
- **Winner, DivHacks 2026** – Won MLH Best Use of Backboard at Columbia University for PlaNYt, an AI NYC trip planner with live group itinerary re-optimization.